Junseo Lee

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Education

Harvard University – *Harvard Kenneth C. Griffin Graduate School of Arts and Sciences* — Cambridge, MA, USA
Doctor of Philosophy (Ph.D.), Quantum Science and Engineering — Starting Fall 2026

- Advisors: Prof. Anurag Anshu and Prof. Sitan Chen (Computer Science), Prof. Jordan Cotler (Theoretical Physics)
- Affiliated with the Harvard Quantum Initiative and the Theory of Computation Group

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
Cross-Registered Graduate Student — Starting Fall 2026

Yonsei University — Seoul, Korea
Bachelor of Science (B.S.), Electrical and Electronic Engineering — Mar. 2019 – Feb. 2023

- Fully funded by the Hyundai Motor CMK Science and Technology Scholarship
- Thesis: *Combinatorial Designs in Information Theory* (Advisor: Prof. Hong-Yeop Song)

Chungnam Science High School — Gongju, Korea
Concentration in Mathematics, Early Honors Graduate — Mar. 2017 – Dec. 2018

Research Positions

Institute of Computer Technology, Seoul National University — Seoul, Korea
Research Associate (Host: Dr. Kabgyun Jeong) — Apr. 2026 – Aug. 2026

Research Institute of Mathematics, Seoul National University — Seoul, Korea
Research Affiliate (Host: Dr. Kabgyun Jeong) — Jan. 2023 – Mar. 2026
Undergraduate Research Assistant (Advisor: Dr. Kabgyun Jeong) — Mar. 2020 – Dec. 2022

Republic of Korea Army — Seoul, Korea
Research Scientist (Technical Research Personnel, Mandatory Military Service) — Mar. 2023 – Mar. 2026

School of Electrical and Electronic Engineering, Yonsei University — Seoul, Korea
Undergraduate Research Assistant (Advisor: Prof. Chulhee Lee) — Jul. 2022 – Dec. 2022

School of Mathematics and Computing, Yonsei University — Seoul, Korea
Undergraduate Research Assistant (Advisor: Prof. Jeehyun Lee) — Dec. 2021 – Feb. 2022

Publications (Google Scholar Profile)

*Equal contribution. †Authors listed alphabetically.

Preprints

(17) Kabgyun Jeong, Donghwa Ji, **Junseo Lee**†, Adam Sawicki, and Oskar Słowik, "Explicit bounds on polylogarithmic spectral gap decay in unitary channels," (2026). In preparation.

(16) Myeongjin Shin, **Junseo Lee**, and Changhun Oh, "Heisenberg-limited Hamiltonian learning without short-time control," (2026). To appear.

(15) Andreas Bluhm, Matthias C. Caro, Francisco Escudero Gutiérrez, **Junseo Lee**†, Aadil Oufkir, Cambyse Rouzé, and Myeongjin Shin, "Certifying and learning local quantum Hamiltonians," (2026). In: *21st Conference on the Theory of Quantum Computation, Communication and Cryptography* (**TQC 2026**, Contributed talk). `arXiv:2603.29809`.

(14) Marco Fanizza, Vishnu Iyer, **Junseo Lee**†, Antonio Anna Mele, and Francesco Anna Mele, "Efficient learning of bosonic Gaussian unitaries," (2025). In: *29th Annual Conference on Quantum Information Processing* (**QIP 2026**, Contributed talk). `arXiv:2510.05531`.

(13) Kartik Anand, Kabgyun Jeong, and **Junseo Lee**†, "Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy," (2025). arXiv:2506.19792.

Journal Articles

(12) Donghwa Ji, **Junseo Lee**, Myeongjin Shin, IlKwon Sohn, and Kabgyun Jeong, "Bounding quantum uncommon information with quantum neural estimators," *Quantum Science and Technology* **11**, 015001 (2026). DOI: 10.1088/2058-9565/ae18f4.

(11) Myeongjin Shin*, **Junseo Lee***, Seungwoo Lee, and Kabgyun Jeong, "Resource-efficient algorithm for estimating the trace of quantum state powers," *Quantum* **9**, 1832 (2025). DOI: 10.22331/q-2025-08-27-1832.

(10) Mingyu Lee, Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, "Mutual information maximizing quantum generative adversarial networks," *Scientific Reports* **15**, 32835 (2025). DOI: 10.1038/s41598-025-18476-y.

(9) Myeongjin Shin*, Seungwoo Lee*, **Junseo Lee***, Donghwa Ji, Hyeonjun Yeo, and Kabgyun Jeong, "Disentanglement provides a unified estimation for quantum entropies and distance measures," *Physical Review A* **110**, 062418 (2024). DOI: 10.1103/PhysRevA.110.062418.

(8) Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, "Estimating quantum mutual information through a quantum neural network," *Quantum Information Processing* **23**, 57 (2024). DOI: 10.1007/s11128-023-04253-1.

(7) **Junseo Lee** and Kabgyun Jeong, "Quantum Rényi entropy functionals for bosonic gaussian systems," *Physics Letters A* **490**, 129183 (2023). DOI: 10.1016/j.physleta.2023.129183

(6) **Junseo Lee**, Hyeonjun Yeo, and Kabgyun Jeong, "Weighted p-Rényi entropy power inequality: Information theory to quantum Shannon theory," *International Journal of Theoretical Physics* **62**, 253 (2023). DOI: 10.1007/s10773-023-05512-8

(5) **Junseo Lee** and Kabgyun Jeong, "High-dimensional private quantum channels and regular polytopes," *Communications in Physics* **31**, 189 (2021). DOI: 10.15625/0868-3166/15762

(4) Kabgyun Jeong, **Junseo Lee**, *et al.*, "Single qubit private quantum channels and 3-dimensional regular polyhedra," *New Physics: Sae Mulli* **68**, 232 (2018). DOI: 10.3938/NPSM.68.232

Book Chapters

(3) **Junseo Lee**, "Assessing Quantum Integer Factorization Performance with Shor's Algorithm," In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*, Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa, *CRC Press* (2024). DOI: 10.1201/9781003475286

Patents

(2) Kabgyun Jeong, Myeongjin Shin, and **Junseo Lee**, "Method for estimating quantum mutual information through a quantum neural network," *Korea Patent* Open No. 10-2026-0009068 (2024). DOI: 10.8080/1020240091151

Notes

(1) **Junseo Lee**† and Myeongjin Shin, "Optimal certification of constant-local Hamiltonians," (2025). arXiv:2512.09778. (This note has been subsumed by a subsequent work; see arXiv:2603.29809 for details.)

Professional Activities

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Research:** npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik

Conference Reviewer

- IEEE Symposium on Foundations of Computer Science (FOCS), 2026
- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Organizing

- *Organizer*, QISCA Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems, 2026
- *Co-organizer*, SNU Quantum Information Theory Seminar, 2024–2025
- *Co-organizer*, Quantum AI Hackathon, jointly organized by Kakao Enterprise Corp. and Jeonju University, 2025

Committee and Reviewing

- *Poster Session Judge*, National Undergraduate Quantum Conference, Seoul National University, 2026
- *Selection Committee*, Quantum Internship Program, National Information Society Agency and Korea Quantum Industry Center, 2024–2025

Outreach and Community Resources

- *Creator and Maintainer*, Quantum Learning Theory Zoo, 2025–
- *Facilitator (Mentor)*, Mathematics Section, Korea Scholar's Conference for Youth (KSCY), Yonsei University, 2019

Selected Honors and Awards

Funding and Fellowships

- *Harvard Graduate Fellowship*, Harvard University, 2026
- *Academic Travel Grant*, Hyundai Motor CMK Foundation, 2022
- *Hyundai Motor CMK Science and Technology Scholarship*, Hyundai Motor CMK Foundation, 2021–2022
- *Teaching Fellowship for Software Courses*, Yonsei University, 2021–2022

Academic Excellence Awards

- *High Honor Student*, Yonsei University, 2022
- *Honor Student*, Yonsei University, 2020–2021
- *JH Science Scholarship*, JH Foundation, 2017
- *Academic Excellence Scholarship*, Asan Future Scholarship Foundation, 2016

Additional Honors and Awards

- *Selected Paper Award*, Finance and Economics Contest, DB Group, 2022
- *Best Tutor Award*, Yonsei University, 2021–2022
- *Third Prize*, Undergraduate Research Exhibition, Korean Physical Society, 2021
- *Bronze Award*, The Humantech Paper Award, Samsung Electronics, 2018
- *Best Translator Award*, NAVER Connect Foundation and Khan Academy, 2018
- *National Honorable Mention and Regional Gold Award*, Korean Olympiad in Informatics, 2016

Research Visits

- Harvard University, hosted by Anurag Anshu and the Harvard Quantum Initiative, Mar. 2026

Teaching

Undergraduate Research Mentoring

(Mentored undergraduate students during appointments at Seoul National University)

- Arul Rhik Mazumder (Carnegie Mellon University), Spring 2026–present

- Kartik Anand (Indian Institute of Technology Goa) [arXiv '25], Summer 2025
- Donghwa Ji (Seoul National University) [Quantum Sci. Technol. '26], Spring 2025–present (co-supervised with Dr. Kabgyun Jeong)
- Mingyu Lee (Seoul National University; Research Intern at University of Oxford) [Sci. Rep. '25], Spring 2024–present (co-supervised with Dr. Sathyawageeswar Subramanian and Dr. Kabgyun Jeong)
- Myeongjin Shin (Korea Advanced Institute of Science and Technology) [PRA '24, Quantum '25, TQC '26], Spring 2023–present (co-supervised with Dr. Kabgyun Jeong)

Quantum Information Science Club Association
(Teaching materials are available at: harris-junseo-lee.github.io/teaching/)

- *Organizer and Lecturer*, Quantum Learning Theory for Bosonic and Fermionic Systems, Lecture 6: "Learning Bosonic Gaussian Unitaries", Winter 2026
- *Invited Lecturer*, Quantum Complexity Reading Group, Fall 2025
- *Invited Lecturer*, Quantum Learning and Complexity Theory, Summer 2025

University–Industry Research Internship

- *Instructor*, AAA558/AAA559: College of Informatics Internship, Korea University (Graduate Course), Fall 2025
- *Instructor*, SW4343: Software Field Placement 1, Korea Aerospace University, Fall 2024

Yonsei University

- *Teaching Assistant*, YCS1009: Change the World through Programming, Fall 2022
- *Teaching Assistant*, YCS1002: Software Programming, Fall 2022
- *Teaching Assistant*, EEE1108: Engineering Information Processing, Fall 2021
- *Course Tutor*, MAT2016: Engineering Mathematics 3, Spring 2022, Best Tutor Award
- *Course Tutor*, MAT1012: Engineering Mathematics 2, Fall 2021, Best Tutor Award

Selected Talks

*Online talk.

Research Talks

"Certifying and learning local quantum Hamiltonians"

- *Contributed talk*, **TQC 2026**, Aug. 2026

"Optimal certification of constant-local Hamiltonians"

- *Invited talk*, Quantum Software Lab Seminar, University of Edinburgh, Mar. 2026*
- *Contributed talk*, Annual Meeting of the Quantum Information Society of Korea, Feb. 2026

"Efficient learning of bosonic Gaussian unitaries"

- *Invited talk*, Annual Meeting of the Quantum Information Society of Korea, Feb. 2026
- *Invited talk*, N^3etFraST Workshop, Nov. 2025
- *Invited talk*, Yonsei Quantum Data Science & AI Lab Seminar, Nov. 2025
- *Contributed talk*, **QIP 2026**, Jan. 2026 (Presented under the title "Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators", as a merged submission with arXiv:2504.11318.)

"Resource-efficient algorithm for estimating the trace of quantum state powers"

- *Invited talk*, Electronics & Telecommunications Research Institute, Dec. 2024
- *Invited talk*, SNU Quantum Information Theory Seminar, Dec. 2024*
- *Invited talk*, IBM-Yonsei Qiskit Fall Fest, Nov. 2024*
- *Contributed talk*, Annual Meeting of Korean Mathematical Society, Oct. 2024
- *Poster*, QIP 2025, Feb. 2025

"Mutual information maximizing quantum generative adversarial network"

- *Invited talk*, Triangle Quantum Computing Seminar, NC State University Quantum Initiative, Nov. 2023*

"Estimating quantum mutual information through a quantum neural network"

- *Invited talk*, CS Katha Barta, National Institute of Science Education and Research Bhubaneswar, Aug. 2023*

"Quantum Rényi entropy functionals for bosonic Gaussian systems"

- *Poster*, QIP 2022, Mar. 2022

"High-dimensional private quantum channels and regular polytopes"

- *Invited talk*, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- *Invited talk*, SNU Quantum Information Theory Seminar, Aug. 2021*
- *Contributed talk*, Winter Meeting of the Optical Society of Korea, Feb. 2022
- *Contributed talk*, Fall Meeting of the Korean Physical Society, Oct. 2021*
- *Poster*, QIP 2022, Mar. 2022

Tutorials and Public Lectures

"Learning theory in ∞-dimensional quantum systems"

- *Invited talk*, Team QST Summer Workshop, Seoul National University, Aug. 2025

"Introduction to quantum machine learning"

- *Invited talk*, AWS Korea Healthcare & Research Team Seminar, Mar. 2025

"Topics in theoretical quantum computer science"

- *Invited talk*, Shinil High School, Aug. 2024

"Quantum machine learning models for drug library generation"

- *Invited talk*, Yonsei Quantum Computing and Monte Carlo Workshop, Aug. 2024

"$\mathsf{QMA} \stackrel{?}{=} \mathsf{NP}$: The NLTS theorem and the quantum PCP conjecture"

- *Invited talk*, SNU Center for Quantum Network's Channel Capacity Summer Workshop, Jul. 2024

"Minimal data may be sufficient for quantum artificial intelligence"

- *Invited talk*, SNU Department of Mathematical Sciences Seminar, Jun. 2023*